

MINGZHE ZHANG

RICHIE | AI APPLICATION DEVELOPER

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Portfolio: babyzmz.github.io/mingzhe-portfolio-v2

GitHub: github.com/babyzmz | English; Mandarin (native)

PROFILE

AI application developer and RMIT Business Information Systems graduate with hands-on experience across desktop and web projects. Work includes model/API integration, document retrieval, tool workflows, file previews and persistent task-state handling. Use Codex and Claude with repository context, code review, debugging and targeted tests to turn requirements into working software.

TECHNICAL SKILLS

Application development TypeScript/JavaScript, Python, React, HTML/CSS, Electron, Tauri/Rust.

Backend & data Node.js, REST APIs, server-sent events (SSE), PostgreSQL/SQLite, SQL.

Applied AI Model APIs, MCP tools, reusable Skills, retrieval-augmented generation (RAG), speech input/output.

Development workflow Codex, Claude, Cursor, Windsurf; Git/GitHub, Docker; requirements, debugging and focused tests.

SELECTED PROJECT EXPERIENCE

Independent AI application development | Current projects

Fairy | Personal Device AI Assistant

Local-first Windows · In development · React / TypeScript · Tauri / Rust · Python

- Develop a desktop assistant that brings ordinary and project conversations, task activity and generated-file previews into one workspace.
- Work on model-provider integration, MCP tool connections and reusable Skills, alongside document-grounded Q&A and voice input/output.
- Refine the desktop companion's SVG appearance, window layout and hit-testing; reproduce real desktop issues alongside build and test checks.
- Explore real-time companionship with local and cloud model options, considering latency, hardware needs and long-session behaviour; advanced features remain in Beta.

Public source: github.com/babyzmz/Fairy-LLM

Mojo Project Family | Core + Personal & Enterprise Workbenches

Ongoing development · Private repositories · Public case studies in portfolio

MojoCore provides shared execution services for MojoClaw and MojoAX. Fairy has its own Core.

MojoCore | Shared Agent Runtime

Node.js / TypeScript · PostgreSQL · REST / SSE

- Develop shared backend capabilities for persistent AI runs, ordered event replay, cancellation and conversation-state recovery.
- Work on duplicate-request handling, authenticated user boundaries and terminal-state consistency to keep tasks traceable and recoverable.
- Extend command-ledger, project and artifact workflows with focused contract and smoke checks, including restart-recovery paths.

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AI Application Developer | Selected projects & background

SELECTED PROJECT EXPERIENCE — CONTINUED

MojoClaw and MojoAX | Product workbenches connected to MojoCore

MojoClaw | Personal AI SaaS — Web & Desktop

React / TypeScript · Electron · Core APIs · In development

- Develop workspaces for conversations, project tasks and generated-file/HTML previews across web and desktop interfaces.
- Connect the application API layer to MojoCore for run creation, streaming, cancellation and conversation restoration, keeping execution state in the backend.
- Investigate authentication, model-provider and message-ordering issues; use logs and targeted smoke checks to reproduce and verify chat-flow behaviour.

MojoAX | Enterprise AI Workbench

React / TypeScript · Tailwind CSS · Shared Core services · In development

- Translate business-system requirements into task-relevant workspaces and panels, including automatic selection, pinning and split views.
- Integrate enterprise interfaces with shared runtime services; contribute to troubleshooting of task activity, loading and failure states.

LIVE WORK SAMPLE

Systems in Motion | Bilingual 3D Portfolio

TypeScript · Vite · WebGL · GSAP / ScrollTrigger · GitHub Pages

- Direct and iterate an AI-assisted portfolio build, defining visual requirements for a scroll-driven 3D narrative, nine project cases and animated detail views.
- Refine presentation, bilingual content, responsive layouts and motion controls; connect project descriptions to public source links and captured application evidence.

The published site includes GitHub Actions build and browser checks. Development uses adapted components alongside project-specific code.

Live site: babyzmz.github.io/mingzhe-portfolio-v2

EXPERIENCE

Beijing Anxin | Intern

Jan 2023 – Mar 2023

- Supported big-data platform construction and maintenance, with exposure to AI applications for production safety and system monitoring.
- Contributed to frontend and backend development tasks under supervision.

EDUCATION

Bachelor of Business Information Systems

Jul 2022 – Dec 2024

RMIT University | Melbourne, VIC

Diploma of Information Technology

2020 – Jun 2021

RMIT University | Melbourne, VIC

ADDITIONAL FOUNDATIONS

Vue.js, Java; Windows and macOS; Linux fundamentals.